GB87362477f rP3

Herbicide

Bandur®

A pre- and post-emergence herbicide for the control of black-grass, annual meadow-grass, and a range of annual broad-leaved weeds in winter wheat, winter barley, potatoes, field beans, and combining peas.

A suspension concentrate formulation containing 600 g/L aclonifen.

The (COSHH) Control of Substances Hazardous to Health Regulations may apply to the use of this product at work.

Bayer CropScience Ltd, PO Box 1582, Cambridge, CB1 0FE
Telephone: 01223 226500

For 24 hour emergency information contact Bayer CropScience Ltd

Telephone:
0330 678 3382 (24 hr)

National Poisons Information Centre UK: 0344 892 0111 (medical professionals only)

National Poisons Information Centre Dublin: +353 1 809 2166

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BAYER

GROUP 32 HERBICIDE

MAPP 19679 H

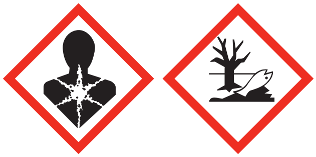
Shake Well

GB87356345f rP3

BANDUR®

UFI: R5T0-G00C-D000-NWT9

A suspension concentrate formulation containing 600 g/L aclonifen.



WARNING

Suspected of causing cancer.
Very toxic to aquatic life with long lasting effects.

Wear protective gloves / protective clothing.
IF exposed or concerned: Get medical advice / attention.
Collect spillage.
Dispose of contents/container to a licensed hazardous-waste disposal contractor or collection site except for empty clean containers which can be disposed of as non-hazardous waste.

Contains aclonifen and 1,2-benzisothiazolin-3-one. May produce an allergic reaction.

To avoid risks to human health and the environment, comply with the instructions for use.

SAFETY PRECAUTIONS

Operator Protection

Engineering control of operator exposure must be used where reasonably practicable in addition to the following personal protective equipment:
WEAR SUITABLE PROTECTIVE GLOVES when handling the concentrate.
However, engineering controls may replace personal protective equipment if a COSHH assessment shows they provide an equal or higher standard of protection.
WHEN USING DO NOT EAT, DRINK OR SMOKE
WASH HANDS AND EXPOSED SKIN before eating and drinking and after work.

Environmental Protection

Do not contaminate water with the product or its container. Do not clean application equipment near surface water. Avoid contamination via drains from farmyards and roads.

To protect aquatic organisms respect an unsprayed buffer zone to surface water bodies in line with LERAP requirements.

To protect aquatic organisms, respect an unsprayed buffer zone to surface water bodies as specified for the crop. HORIZONTAL BOOM SPRAYERS MUST BE FITTED WITH THREE STAR DRIFT REDUCTION TECHNOLOGY. Low drift spraying equipment must be operated according to the specific conditions stated in the official three star rating for that equipment as published on HSE Chemicals Regulation Division's website. Maintain three star operating conditions until 30 m from the top of the bank of any surface water bodies.

DO NOT ALLOW DIRECT SPRAY from horizontal boom sprayers to fall within the distance specified for the crop to the top of the bank of a static or flowing water body, or within 1 m of the top of a ditch which is dry at the time of application. Aim spray away from water. NOTE: BUFFER ZONES OF MORE THAN 5 M CANNOT BE REDUCED UNDER THE LOCAL ENVIRONMENT RISK ASSESSMENT FOR PESTICIDES (LERAP) SCHEME.

The statutory buffer zone must be maintained and the distance recorded in Section A of the LERAP record form. The LERAP record form must be kept available for three years.

Extreme care must be taken to avoid spray drift onto non-crop plants outside of the target area.

Storage and Disposal

KEEP AWAY FROM FOOD, DRINK AND ANIMAL FEEDING STUFFS.
KEEP OUT OF REACH OF CHILDREN.
KEEP IN ORIGINAL CONTAINER, tightly closed, in a safe place.
WASH OUT CONTAINER THOROUGHLY, empty washings into spray tank and dispose of container safely.

To access the **Safety Data Sheet** for this product scan the code or use the link below:
https://cropscience.bayer.co.uk/our-products/herbicides/bandur
or alternatively contact your supplier

Bayer

IMPORTANT INFORMATION FOR USE ONLY AS A PROFESSIONAL HERBICIDE

Crops	Maximum individual dose (L product/ha)	Maximum total dose (L product/ha/ crop)	Maximum number of treatments (per crop)	Latest time of application
Wheat (winter)	1.4	-	1	Pre-emergence (BBCH 00-09)
	OR			
	1.0	1.4	-	Pre-emergence (BBCH 00-09)
	AND/OR			
	0.4	1.4	-	Before the four true leaf stage of the crop
Barley (winter)	1.0	-	1	Pre-emergence (BBCH 00-09)
	OR			
	1.0	1.4	-	Pre-emergence (BBCH 00-09)
	AND/OR			
	0.4	1.4	-	Before the four true leaf stage of the crop
Potato	1.75	-	1	Before crop emergence (not less than 42 days before harvest)
Field beans, combining peas	1.4	-	1	Before crop emergence

Aquatic buffer zone distance: 6 metres

Other specific restrictions: This product must not be applied via hand-held equipment.

Buffer zones greater than 5 m are NOT eligible for buffer zone reduction under the LERAP scheme.

Low drift spraying equipment must be operated according to the specific conditions stated in the official three star rating for that equipment as published on HSE Chemicals Regulation Division's website. These operating conditions must be maintained until the operator is 30 m from the top of the bank of any surface water bodies.

READ THE LABEL BEFORE USE. USING THIS PRODUCT IN A MANNER THAT IS INCONSISTENT WITH THE LABEL MAY BE AN OFFENCE. FOLLOW THE CODE OF PRACTICE FOR USING PLANT PROTECTION PRODUCTS.

DIRECTIONS FOR USE

IMPORTANT: This information is approved as part of the Product Label. All instructions within this section must be read carefully in order to obtain safe and successful use of this product.

Bandur is a diphenylether herbicide for the pre- and post-emergence control of black-grass, annual meadow-grass and annual broad-leaved weeds in winter wheat, winter barley, potatoes, field beans and combining peas.

RESTRICTIONS

DO NOT disrupt the soil surface after application as this will reduce the level weed control provided.

On sandy, stony or gravelly soils there is a risk of crop damage, especially if heavy rain falls soon after application. Avoid applying Bandur when rain is imminent.

Do not use on waterlogged soil or soils prone to waterlogging.

Do not apply to shallow drilled crops.

PROTECT FROM FROST.

HERBICIDE RESISTANCE MANAGEMENT

When herbicides with the same mode of action are used repeatedly over several years in the same field, selection of resistant biotypes can take place. These can propagate and may become dominating. A weed species is considered to be resistant to a herbicide if it survives a correctly-applied treatment at the recommended dose. Strains of some annual grasses, e.g. black-grass, wild-oats, and Italian rye-grass, have developed resistance to herbicides which may lead to poor control. A strategy for preventing and managing such resistance should be adopted. This should include integrating herbicides with a programme of cultural control measures. Guidelines have been produced by the Weed Resistance Action Group and copies are available from the AHDB, CropLife UK, your distributor, crop adviser or product manufacturer.

Key aspects of the Bandur Resistance Management Strategy are:

- ALWAYS follow WRAG guidelines for preventing and managing herbicide resistant weeds.
- DO NOT use Bandur as a stand-alone treatment for grass weed control. Use only in tank mix or sequence with effective herbicides with alternative modes of action.
- DO NOT use Bandur as the sole means of grass weed or broad-leaved weed control in successive crops.
- ALWAYS use grass and broad-leaved weed herbicides with alternative modes of action throughout the cropping rotation.
- ALWAYS monitor weed control effectiveness and investigate any odd patches of poor grass or broad-leaved weed control. If unexplained contact your agronomist who may consider a resistance test appropriate.

WEEDS CONTROLLED

Bandur is absorbed primarily through the shoot of emerging seedlings as they grow through the layer of herbicide applied pre-emergence to the soil surface. Susceptible weeds can emerge, but these will become chlorotic, their growth will be retarded leading ultimately to death. After pre-emergence application do not disrupt the herbicide layer. Any form of mechanical cultivation or disruption of the soil surface by any other method will reduce the level of weed control provided.

Weed susceptibility ratings in the tables below are for applications made pre-emergence of the weeds, unless states otherwise.:

Winter wheat and winter barley

Weed	Susceptibility at 1.0 L/ha pre-emergence	Susceptibility at 1.4 L/ha pre-emergence (winter wheat only)	Susceptibility at 0.4 L/ha post-emergence	1.0 L/ha pre-emergence and 0.4 L/ha post-emergence
Black-grass	-	Pre-emergence (MR) and post-emergence to the two-leaf stage (provides some control)	-	-
Annual meadow-grass	Pre-emergence (S)	-	Post-emergence to the two-true leaf stage (MR)	Post-emergence to the two-true leaf stage (S)
Cleavers	-	Pre-emergence (MR) and post-emergence to the two leaf stage (MR)	-	-
Mayweed, scented	Pre-emergence (MR)	Pre-emergence (MS) and post-emergence to one-true leaf (MR)	Post-emergence to the one-true leaf stage (MR)	-
Field pansy	Pre-emergence (MR)	Pre-emergence (MS)	Post-emergence to the one-true leaf stage (R)	-
Common chickweed	Pre-emergence (S)	-	Post-emergence to the two-true leaf stage (S)	Post-emergence to the two-true leaf stage (S)

R – Resistant MR – Moderately resistant MS – Moderately susceptible
S - Susceptible

*Control of cleavers is reduced in organic soils

Potato

Weed	1.75 L/ha
Cleavers*	MR
Pansy, field	MS
Bindweed, black	MR
Fat-hen	S
Mayweeds	S
Redshank	MS

R – Resistant MR – Moderately resistant MS – Moderately susceptible
S - Susceptible

*Control of cleavers is reduced in organic soils

Field beans and combining peas

Weed	Winter beans 1.4 L/ha	Winter beans 1.0 L/ha	Spring beans and spring peas 1.4 L/ha
Black-grass	Pre-emergence MR & post-emergence to the two-leaf stage (provides some control)	-	-
Cleavers*	Pre-emergence MR and post-emergence to the two-leaf stage (MR)	-	-
Pansy, field	MS	MR	MS
Bindweed, black	-	-	MR
Fat-hen	-	-	S
Mayweeds	-	-	MS
Mayweed, scented	Pre-emergence MS & post-emergence to one-true leaf (MR)	MR	-
Redshank	-	-	MS

R – Resistant MR – Moderately resistant MS – Moderately susceptible
S - Susceptible

*Control of cleavers is reduced in organic soils

To avoid increased selection of herbicide-resistant black-grass, always use Bandur in tank mixture with a suitable partner which has a different mode of action, and which is effective against black-grass.

Established broad-leaved weeds growing from rootstocks will not be controlled by Bandur. As for other residual herbicides application to a fine, firm seedbed will optimise efficacy. Efficacy will be reduced where application is made to cloddy seedbeds or where there is disruption of the soil surface after application. Effectiveness using three star drift reduction technology may be reduced.

CROP SPECIFIC INFORMATION

Winter wheat and winter barley

Good weed control depends on burying any trash or straw before or during seedbed preparation.

Seedbeds must have a firm, fine tilth. Loose or cloddy seed beds must be consolidated otherwise crop damage may result due to inadequate seed cover. For pre-emergence treatments, seed should be covered with a minimum of 32 mm of settled soil.

Bandur may be applied either pre-emergence of the crop, post-emergence before the four true leaf stage, or as a sequential application pre- and post-emergence.

Apply Bandur pre-emergence of the crop, at a maximum rate of 1.4 litres of product per hectare on winter wheat, or 1.0 litres of product per hectare on winter barley.

A sequential application of 1.0 L/ha pre-emergence followed by a 0.4 L/ha post-emergence may provide a useful contribution to the residual control of annual meadow grass and common chickweed in winter wheat and winter barley, particularly in situations where germination is protracted and emergence after application is anticipated. Do not apply more than 0.4 L/ha post-emergence of the crop and do not exceed the maximum total dose of 1.4 L/ha.

Apply via a horizontal boom sprayer, in 200 to 400 litres of water per hectare as a **MEDIUM** spray (BCPC category). A spray pressure of at least 2 bars is advised. Good, even spray coverage of soil and weeds is essential. Ensure that spray swaths do not overlap. To prevent damage, care must be taken to avoid drift onto neighbouring crops.

Cultivation after spraying will encourage weed germination and reduce the residual activity of Bandur. On mineral soils with a high organic matter content and on peaty or organic soils the residual activity of Bandur may be reduced.

Potato

Bandur may be used on all commercial varieties of potatoes

Apply Bandur at a maximum rate of 1.75 litres of product per hectare. Application must be pre-emergence of the crop and weeds. Apply via a horizontal boom sprayer, in 200 to 400 litres of water per hectare as a **MEDIUM** spray (BCPC category).

Ridges should be well rounded with few clods. Cultivations should produce a soil tilth that requires no further improvement after planting. Cultivation after spraying will encourage weed germination and reduce the residual activity of Bandur. On mineral soils with a high organic matter content and on peaty or organic soils the residual activity of Bandur may be reduced.

Ensure Bandur is applied evenly overall to both sides of potato ridges. Care must be taken to ensure that application takes place before crop emergence.

Only one application of Bandur should be made to the crop.

Field beans and combining peas
Bandur may be used on all spring and winter varieties of field beans (Vicia faba).

Apply Bandur at a maximum rate of 1.4 litres of product per hectare. Application must be pre-emergence of the crop and weeds, although some control of weeds treated post-emergence may be achieved. Apply via a horizontal boom sprayer, in 200 to 400 litres of water per hectare as a **MEDIUM** spray (BCPC category). Good, even spray coverage of soil and weeds is essential. Ensure that spray swaths do not overlap.

Only one application of Bandur should be made to the crop.

FOLLOWING CROPS and CROP FAILURE

Following cultivation to 10 cm, winter wheat, winter barley, winter triticale, winter rye, winter oats, winter oilseed rape and winter field pea may be sown in the year of harvest to succeed a winter cereal crop treated with Bandur. Spring oilseed rape, sugar beet, spring field pea, sunflower, dwarf french bean, spring wheat, spring barley, spring oats and maize may be drilled in the spring following harvest of the winter cereal crop treated with Bandur.

In the event of failure of the treated crop, provided that cultivation and thorough mixing of treated soil has been conducted to at least 10 cm depth any of the above listed crops can be established in the appropriate season where at least 20 days has elapsed after application of Bandur to the cereal crop.

After a potato, pea or bean crop treated with Bandur, oilseed rape, combining pea, dwarf french bean, wheat, barley, oat, winter triticale, winter rye, sugar beet, maize, and sunflower can be established in the normal rotation or in the event of failure of the treated crop, provided that cultivation (using cultivator, disc harrow or similar) and thorough mixing of treated soil has been conducted to at least 10 cm depth.

Radish and phacelia require a period of at least 120 days after application, or plough to at least 20 cm before drilling.

MIXING

Shake the container well before use. Half fill the spray tank with clean water, begin agitation and add the required amount of Bandur. Wash out the container and add the washings to the spray solution, before topping up with clean water. Continue agitation until spraying is completed. Do not leave the sprayer filled with the spray solution standing for long periods. Wash out the sprayer thoroughly after use using a wetting agent or proprietary tank cleaner.